# Proofs and further applications of polynomial identities

**ANSWER KEY**

## Core Concepts: Proofs and further applications of polynomial identities

**1** For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(2)$.

**a.** $17$ ✓

**b.** $21$

**c.** $8$

**d.** $21$

Substitute $x = 2$: $f(2) = 4(2) + 9 = 17$.

**2** Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

**a.** $x = 1$ and $x = 7$

**b.** $x = 1$ **and** $x = 6$ ✓

**c.** $x = -1$ and $x = -6$

**d.** $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

**3** A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** $14.5$

**b.** $15.5$

**c.** $16.5$ ✓

**d.** $18.5$

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by $4$ gives $16.5$.

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## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(3)$.

**a.** $21$ ✓

**b.** $25$

**c.** $12$

**d.** $25$

Substitute $x = 3$: $f(3) = 4(3) + 9 = 21$.

**5** Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

**a.** $x = 2$ and $x = 8$

**b.** $x = 2$ **and** $x = 7$ ✓

**c.** $x = -2$ and $x = -7$

**d.** $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

**6** A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** $14.5$

**b.** $15.5$

**c.** $16.5$ ✓

**d.** $18.5$

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by $4$ gives $16.5$.

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## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(4)$.

**a.** $25$ ✓

**b.** $29$

**c.** $16$

**d.** $29$

Substitute $x = 4$: $f(4) = 4(4) + 9 = 25$.

**8** Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ **and** $x = 8$ ✓

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

**9** A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** $14.5$

**b.** $15.5$

**c.** $16.5$ ✓

**d.** $18.5$

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by $4$ gives $16.5$.

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## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 4x + 9$ in proofs and further applications of polynomial identities, evaluate $f(5)$.

**a.** $29$ ✓

**b.** $33$

**c.** $20$

**d.** $33$

Substitute $x = 5$: $f(5) = 4(5) + 9 = 29$.

**11** Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

**a.** $x = 4$ and $x = 10$

**b.** $x = 4$ **and** $x = 9$ ✓

**c.** $x = -4$ and $x = -9$

**d.** $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

**12** A data set recorded for proofs and further applications of polynomial identities consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** $14.5$

**b.** $15.5$

**c.** $16.5$ ✓

**d.** $18.5$

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by $4$ gives $16.5$.

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